How to find $E[g(X)]$?

let $Y = g(X)$

Suppose Range of $Y = g(X) = \{y_1, y_2\}$

$$E[g(X)] = \sum_{y:\, y \in \text{Range}(g(X))} y\, P(Y=y)$$

$$= \sum_{y} y \underbrace{\boxed{\sum_{\substack{x: \\ g(x)=y}} P(X=x)}}_{P(Y=y)}$$

$x_1, x_2 \to y_1$

$x_3, x_4 \to y_2$

$$= \underbrace{y_1\left(P(X=x_1) + P(X=x_2)\right)}_{g(x_1)=y_1=g(x_2)} + y_2\left(P(X=x_3) + P(X=x_4)\right)$$

$g(x_3) = g(x_4) = y_2$

Range of $X = \{x_1, x_2, x_3, x_4\}$

$$\boxed{E[g(X)] = \sum_{x:\, P(X=x)>0} g(x)\, P(X=x)}$$

LOTUS

$E[X - E[X]] = 0$

Variance of a r.v. X

$$Var(X) = E[(X - E[X])^2]$$

Standard deviation $(X) = \sqrt{Var\, X}$

On expanding

$$Var(X) = E(X^2) - (E(X))^2$$

Properties of Expectation & variance

1. $E(X+Y) = E(X) + E(Y)$
2. $E(aX+b) = aE(X) + E(b)$
   $= aE(X) + b$

3. $E[g(X)] = \sum_{x} g(x) P(X = x)$

Variance

1. $Var(X + c) = Var(X)$

$E(((X+c) - E(X+c))^2)$

$E((X - E(X))^2) = Var(X)$

2. $Var(cX) = c^2 Var(X)$

$E[(cX - E[cX])^2] = E[c^2 (X - E[X])^2]$

$(c(X - E(X)))^2 \quad = c^2 Var(X)$

Prob: Are there discrete random variable X & Y such that $E(X) > 10^2 E(Y)$ but

$P(Y > X) \geq 0.99$

Sol^n: $P(X = 0) = 0.99$; $P(Y = 1) = 1$

$P(X = 10001) = 0.01$

$E(X) = 10001 \times 0.01 + 0 \times 0.99$

$E(X) > 10^2 E(Y)$

$E(Y) = 1$

$\boxed{\text{Bernoulli} : \; X \sim Ber(p)}$

PMF : $P(X=1) = p, \quad P(X=0) = q$

$E(X) = p.1 + 0.q = p$

$Var(X) = E(X^2) - (E(X))^2$

$E(X) = p - p^2 = p(1-p) = pq$

$\boxed{\text{Binomial} : \; X \sim Bin(n, p)}$

PMF : $P(X=k) = {}^nC_k \, p^k q^{n-k}$

$\{k = 0, 1, 2, \ldots, n\}$

$E(X) = np$

$X = X_1 + X_2 + \ldots + X_n$

$X_i \sim Ber(p)$

$Var(X) =$

$Var(X) = Var(X_1 + X_2 + \ldots + X_n)$

$= Var(X_1) + Var(X_2) + \ldots + Var(X_n)$

If $Var(X+Y) = Var(X) + Var(Y)$ $\rightarrow$ X, Y are independent r.v.s

$= npq$

Ex: To find $Var(X)$ using the formula

$Var(X)$

$\boxed{\text{Geometric} : X \sim Geom(p)}$

imp $\{ X = \#$ of trials to get 1st success

PMF : $P(X=k) = (1-p)^{k-1} p$

$$E(X) = \sum_{k=1}^{\infty} k P(X=k)$$

$x^2 = x(x-1) + x$

$$= p \sum_{k=1}^{\infty} k q^{k-1} = \frac{1}{p}$$

$\sum k^2 \cdot P(X=k)$

$$Var(X) = E[X^2] - (E[X])^2$$

$$= E[X(X-1)] + E[X] - (E[X])^2$$

$$E[X(X-1)] = p \sum_{k=2}^{\infty} \underline{k(k-1) q^{k-1}}$$

$$= pq \cdot \underbrace{\sum_{k=2}^{\infty} k(k-1) q^{k-2}}$$

$$= pq \cdot \frac{2}{(1-q)^3} = \frac{2pq}{p^3} = \frac{2q}{p^2}$$

Aside: $\frac{1}{1-q} = \sum_{k=0}^{\infty} q^k \Rightarrow \frac{1}{(1-q)^2} = \sum_{k=1}^{\infty} k q^{k-1}$

$$\frac{2}{(1-q)^3} = \sum_{k=2}^{\infty} k(k-1)q^{k-2} \quad \text{—— } *$$

$$\Rightarrow \quad E[X(X-1)] = p\sum_{k=2}^{\infty} k(k-1)q^{k-1}$$

$$= pq\sum_{k=2}^{\infty} k(k-1)q^{k-2}$$

From $(*)$

$$= pq\cdot\frac{2}{(1-q)^3} = \frac{2pq}{(1-q)^3} = \frac{2pq}{p^3} = \frac{2q}{p^2}$$

$$Var(X) = E[X(X-1)] + E[X] - (E[X])^2$$

$$= \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} =$$

$$= \frac{2q}{p^2} + \frac{1}{p} - \frac{2}{p^2} + \frac{1}{p^2} = \frac{2(q-1)}{p^2} + \frac{1}{p} + \frac{1}{p^2}$$

$$= -\frac{2p}{p^2} + \frac{1}{p} + \frac{1}{p^2} = -\frac{1}{p} + \frac{1}{p^2}$$

$$= \frac{-p+1}{p^2} = \frac{q}{p^2}$$

Similarly, if

Y = # failures before the 1st success

$Y = X - 1$

$$E[Y] = E[X-1] = E[X] - 1 = \frac{1}{p} - 1 = \frac{q}{p}$$

$$Var[Y] = Var[X-1]$$
$$= Var[X] = \frac{q}{p^2}$$

$\boxed{\text{Negative Binomial: } X \sim NBin(r, p)}$

X = # of failures until rth success

PMF: $P(X=k) = {}^{k+r-1}C_k \cdot (1-p)^k p^r$

$k \in \{0, 1, 2, 3 \ldots$

$$E[X] = \sum_{k=0}^{\infty} k P(X=k) \quad \text{Tedious}$$

Instead,

recall $X = Y_1 + Y_2 + \cdots + Y_r$

Y_i = # failures between $(i-1)$th & ith success

$E[Y_i] = q/p$ (from above)

$E[X] = rq/p$

$$Var(X) = \frac{rq}{p^2}$$

$\boxed{\text{Hypergeometric: } X \sim HGeom(w, b, n)}$

X = # of white balls drawn without replacement from a sample of n balls

PMF of X: $P(X=k) = \dfrac{{}^{w}C_k \cdot {}^{b}C_{n-k}}{{}^{w+b}C_n}$

$0 \le k \le w$

$$E[X] = \sum_{k=0}^{w} k\, P(X=k)$$

$$j\, {}^{n}C_j = n\, {}^{n-1}C_{j-1}$$

$$= \sum_{k=0}^{\infty} \frac{k\, {}^{w}C_k \cdot {}^{b}C_{n-k}}{{}^{w+b}C_n}$$

$$n \cdot \frac{{}^{w+b}C_n}{} = w+b \cdot {}^{w+b-1}C_{n-1}$$

$$= \sum_{k} w \frac{{}^{w-1}C_{k-1} \cdot {}^{b}C_{n-k}}{{}^{w+b}C_n}$$

$$= \frac{wn}{w+b} \sum_{k} \frac{{}^{w-1}C_{k-1} \cdot {}^{b}C_{n-k}}{{}^{w+b-1}C_{n-1}}$$

$$\sum P(X=i) = 1$$

$HGeom(w-1, b, n-1)$

let $i = k-1$

$$= \frac{wn}{w+b} \sum_{i=0}^{w-1} \frac{{}^{w-1}C_i \cdot {}^{b}C_{n-(i+1)}}{{}^{w+b-1}C_{n-1}} = 1$$

$$= \frac{wn}{w+b}$$

To find variance $Var(X)$
compute $E[X(X-1)]$

(Hint : Use $k(k-1)\,{}^{n}C_{k} = n(k-1)\,{}^{n-1}C_{k-1}$
$= n(n-1)\,{}^{n-2}C_{k-2}$)

$\boxed{\text{Poisson Distribution : } X \sim Pois(\lambda)}$

PMF of $X = P(X=k) = \dfrac{e^{-\lambda}\lambda^{k}}{k!}$; $k=0,1,2..$

$$E(X) = e^{-\lambda}\sum_{k=0}^{\infty}\frac{k\lambda^{k}}{k!} = \lambda e^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!}$$

$$\Rightarrow \boxed{E[X] = \lambda e^{-\lambda}e^{\lambda} = \lambda}$$

For Variance : $Var(X) = E[X(X-1)] + E[X] - (E[X])^{2}$

$$E[X(X-1)] = e^{-\lambda}\sum_{k=2}^{\infty}\frac{k(k-1)\lambda^{k}}{k!}$$

$$= e^{-\lambda}\sum_{k=2}^{\infty}\frac{\lambda^{k}}{(k-2)!}$$

$k-2=i$

$$= e^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^{2+i}}{i!}$$

$$= \lambda^2 e^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

$$= \lambda^2 e^{-\lambda} \cdot e^{\lambda} = \lambda^2$$

$$Var(X) = E[X(X-1)] + E[X] - (E[X])^2$$

$$= \lambda^2 + \lambda - \lambda^2 = \lambda$$

$$\boxed{Var(X) = \lambda}$$